

The Gabor System $\mathcal{G}(Q_2, 8/21, 7/4)$ Is a Frame

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190
xiaojianzeng@foxmail.com

Abstract

For the second-order cardinal B-spline $Q_2(x) = (1 - |x|)_+$, we prove that $\mathcal{G}(Q_2, 8/21, 7/4)$ is a frame for $L^2(\mathbb{R})$. The density is $ab = 2/3$, so this is a fixed $p = 2$ rational-density point, outside both the $ab = 1/q$ classification and the applicable range of the nearest known obstruction. We reduce the question to full row rank of a 2×3 Zibulski–Zeevi matrix. An exact certificate partitions one Zak period at ten rational breakpoints, reduces the three maximal minors to polynomials in x and the squared phase, and excludes their simultaneous vanishing by rational Bernstein-sign and polynomial-gcd calculations. The conclusion is only for the displayed parameter pair; no neighborhood or general converse for densities p/q with $p > 1$ is asserted.

1 Context and statement

For $g \in L^2(\mathbb{R})$ and $a, b > 0$, write

$$\mathcal{G}(g, a, b) = \{e^{2\pi i b m x} g(x - a k) : m, k \in \mathbb{Z}\}.$$

The recent analysis of Stangl and Frederick [2] characterizes the relevant frame-set question along $ab = 1/q$ (for the range treated there) and recalls the Lemvig–Nielsen obstruction: if $b > 3/2$, $ab = p/q < 1$ in lowest terms, and $|b - \lfloor b + \frac{1}{2} \rfloor| \leq 1/(2q)$, then the Q_2 system is not a frame [1]. At the point considered here,

$$(a, b) = \left(\frac{8}{21}, \frac{7}{4}\right), \quad ab = \frac{2}{3}, \quad \left|\frac{7}{4} - 2\right| = \frac{1}{4} > \frac{1}{6}.$$

Thus the obstruction does not decide this point, while the $ab = 1/q$ result has numerator $p = 1$ and likewise does not cover it. Our claim is the following fixed-parameter resolution.

Theorem 1. *Let $Q_2(x) = \max(1 - |x|, 0)$. There are constants $0 < A \leq B < \infty$ such that every $f \in L^2(\mathbb{R})$ satisfies*

$$A\|f\|_2^2 \leq \sum_{m, k \in \mathbb{Z}} \left| \left\langle f, e^{2\pi i (7/4)m \cdot} Q_2\left(\cdot - \frac{8}{21}k\right) \right\rangle \right|^2 \leq B\|f\|_2^2.$$

The proof supplies an existential uniform lower bound through compactness; it does not claim a sharp or numerically optimized value of A .

2 The rational Zak matrix

For $\lambda > 0$, use the normalized Zak transform

$$Z_\lambda g(u, \gamma) = \sqrt{\lambda} \sum_{j \in \mathbb{Z}} g(\lambda(u - j)) e^{2\pi i j \gamma}.$$

The rational Zibulski–Zeevi characterization [4, 2] says, in the present $p/q = 2/3$ case, that the Gabor frame property is equivalent to uniform positive lower and finite upper bounds for the eigenvalues of $\Phi\Phi^*$, where, up to harmless convention-dependent unitary factors,

$$\Phi(x, \gamma) = \frac{1}{\sqrt{2}} \begin{pmatrix} Z_{4/7}Q_2(x, \gamma) & Z_{4/7}Q_2(x - 2/3, \gamma) & Z_{4/7}Q_2(x - 4/3, \gamma) \\ Z_{4/7}Q_2(x, \gamma + 1/2) & Z_{4/7}Q_2(x - 2/3, \gamma + 1/2) & Z_{4/7}Q_2(x - 4/3, \gamma + 1/2) \end{pmatrix}. \quad (1)$$

It is therefore enough to prove that this matrix has row rank two at every point of a compact fundamental torus. Indeed, its entries are continuous, so pointwise full row rank then gives a strictly positive minimum for the least eigenvalue of $\Phi\Phi^*$; the largest eigenvalue is automatically bounded.

Put $z = e^{2\pi i \gamma}$ and

$$S(u, z) = \sum_{j \in \mathbb{Z}} \left(1 - \frac{4}{7}|u - j|\right)_+ z^j. \quad (2)$$

Since $Z_{4/7}Q_2(u, \gamma) = \sqrt{4/7} S(u, z)$, deleting the common nonzero scalar $\sqrt{2/7}$ from (1) leaves the three columns

$$\begin{pmatrix} S(x - c, z) \\ S(x - c, -z) \end{pmatrix}, \quad c \in \left\{0, \frac{2}{3}, \frac{4}{3}\right\}. \quad (3)$$

For $0 \leq x \leq 1$, the arguments $x - c$ lie in $[-4/3, 1]$. The support condition in (2) consequently permits only $j = -3, -2, -1, 0, 1, 2$. A summand changes support or affine slope exactly when $x - c - j \in \{-7/4, 0, 7/4\}$. Including the endpoints, the complete ordered list of breakpoints is

$$0, \frac{1}{12}, \frac{1}{4}, \frac{1}{3}, \frac{5}{12}, \frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{11}{12}, 1. \quad (4)$$

3 Exact full-rank certificate

Let D_{rs} be the determinant of columns r, s in (3), indexed by $r, s \in \{0, 1, 2\}$. On each of the nine open cells in (4), direct expansion gives

$$D_{rs}(x, z) = \alpha_{rs} z^{n_{rs}} P_{rs}(x, y), \quad y = z^2, \quad (5)$$

where $\alpha_{rs} \in \mathbb{Q} \setminus \{0\}$, $n_{rs} \in \mathbb{Z}$, and $P_{rs} \in \mathbb{Z}[x, y]$ is primitive. The parity reduction in (5) is also immediate from $D_{rs} = S_r(z)S_s(-z) - S_s(z)S_r(-z)$: only one parity of powers of z survives. As $z \neq 0$, the removed factors do not alter the zero set.

We use the following elementary unit-circle test.

Lemma 2. *Suppose $P(y) = Ay^2 + By + C$, with $A, B, C \in \mathbb{R}$ and $A \neq C$. If $|y| = 1$ and $P(y) = 0$, then $y \in \{1, -1\}$.*

Proof. Since $y^{-1} = \bar{y}$,

$$\operatorname{Im}(y^{-1}P(y)) = \operatorname{Im}(Ay + B + Cy^{-1}) = (A - C) \operatorname{Im} y.$$

The left side is zero when $P(y) = 0$. Hence $\operatorname{Im} y = 0$, and a real point on the unit circle is 1 or -1 . $\square$

For each cell, Table 1 selects a minor of degree at most two in y . It gives $A - C$ and its exact degree-two Bernstein coefficients on the closure of that cell. All three controls have one strict sign, so $A - C$ has that sign throughout the closed interval. This avoids any floating-point sign decision.

Table 1: Exact exclusion of nonreal unit phases.

cell I	minor	$A - C$	Bernstein coefficients on $\bar{I}$
$(0, 1/12)$	P_{01}	$48x^2 + 64x + 17$	17, 59/3, 68/3
$(1/12, 1/4)$	P_{01}	$48x^2 + 64x + 17$	68/3, 86/3, 36
$(1/4, 1/3)$	P_{02}	$-(4x - 7)(12x - 1)$	12, 44/3, 17
$(1/3, 5/12)$	P_{02}	$48x^2 + 32x + 1$	17, 59/3, 68/3
$(5/12, 7/12)$	P_{01}	$-48x^2 + 128x - 81$	-36, -86/3, -68/3
$(7/12, 2/3)$	P_{01}	$-48x^2 + 128x - 81$	-68/3, -59/3, -17
$(2/3, 3/4)$	P_{01}	$(4x + 3)(12x - 11)$	-17, -44/3, -12
$(3/4, 11/12)$	P_{02}	$-48x^2 + 160x - 129$	-36, -86/3, -68/3
$(11/12, 1)$	P_{01}	$-(4x - 3)(12x - 29)$	12, 44/3, 17

It remains to exclude $y = \pm 1$. Let

$$g_{\pm}(x) = \gcd_{\mathbb{Q}[x]}(P_{01}(x, \pm 1), P_{02}(x, \pm 1), P_{12}(x, \pm 1)),$$

normalized to be monic. The exact Euclidean-algorithm results are recorded in Table 2. In every row $g_- = 1$. Each nonconstant g_+ has its sole root outside the corresponding open cell. Lemma 2 therefore proves that the three minors cannot vanish simultaneously on any open cell for $|y| = 1$.

Table 2: Exact common factors after the two real unit-phase specializations.

cell I	$g_+(x)$	$g_-(x)$
$(0, 1/12)$	$x - 1/4$	1
$(1/12, 1/4)$	1	1
$(1/4, 1/3)$	$x - 1/12$	1
$(1/3, 5/12)$	$x - 7/12$	1
$(5/12, 7/12)$	1	1
$(7/12, 2/3)$	$x - 5/12$	1
$(2/3, 3/4)$	$x - 11/12$	1
$(3/4, 11/12)$	1	1
$(11/12, 1)$	$x - 3/4$	1

At a breakpoint the affine formula must be evaluated with its actual endpoint weights, including zeros. Direct rational expansion at each of the ten points in (4) gives

$$\gcd_{\mathbb{Q}[y]}(P_{01}, P_{02}, P_{12}) = 1. \quad (6)$$

Here the normalization is performed anew at the boundary; no division by an x -dependent expression is used. Equation (6) rules out a common complex zero, and in particular a common unit-circle zero, at every boundary point.

For reproducibility, the ancillary Python/SymPy program `evidence/zz_exact_certificate.py` constructs (2) directly, checks the completeness of (4), prints all 27 cell polynomials in (5), performs the exact sign and gcd checks in Tables 1–2, and verifies (6). It uses rational polynomial arithmetic only; its final assertion is

PASS: exact rank two on every open cell and every breakpoint.

Exact Bernstein certificates have also been used for compact complex-phase, rank-two matrix reductions in an unrelated setting [3]; that work is cited only for this adjacent computer-assisted methodology, not for a Gabor-frame result.

Proof of Theorem 1. The open-cell calculation and (6) show that at least one of the three maximal minors of (1) is nonzero for every $(x, \gamma) \in [0, 1] \times (\mathbb{R}/\mathbb{Z})$. Thus Φ has row rank two everywhere. Its entries are continuous finite sums, so $\lambda_{\min}(\Phi\Phi^*)$ is a continuous positive function on this compact torus and has a positive minimum. The largest eigenvalue is continuous and has a finite maximum. The rational Zibulski–Zeevi characterization now gives the two frame bounds asserted in the theorem. $\square$

Scope and assistance disclosure

The result decides only $(a, b) = (8/21, 7/4)$. It neither characterizes a neighborhood nor proves a converse for general reduced densities p/q with $p > 1$.

Python 3 and SymPy were used for exact rational Laurent-polynomial expansion, Bernstein-sign verification, and Euclidean polynomial gcds. Generative-AI tools assisted in proof discovery, certificate design, independent exact replay in a separate model context, and preparation of the exposition and LaTeX. The separate replay is an internal verification step, not external peer review. The author reviewed the mathematical statement, the exact certificate, and the manuscript, and takes responsibility for their content.

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